

IVANA SANCHEZ OLIVARES

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EDUCATION

The University of British Columbia (UBC) M.Sc. in Bioinformatics Relevant Coursework: Methods for High-Dimensional Biology, Algorithms for Bioinformatics (Deep Learning-focused) Awards: CONAHCYT Merit-Based Scholarship , covering full tuition, living stipend, and healthcare. GPA: 96.2/100. Supervisor: Dr. Inanc Birol. Thesis: Simulation of Haplotype-Resolved Single-Cell Methylation Profiles.	Vancouver, Canada Sep. 2024 – Aug. 2026
Universidad Nacional Autónoma de México (UNAM) B.Sc. in Applied Mathematics Relevant Coursework: Bayesian Statistics, Multivariate Analysis, Survival Analysis, Stochastic Simulation. Awards: Gabino Barreda Medal for achieving the highest GPA in the graduating cohort. GPA: 10.00/10.00	Mexico City, Mexico Aug. 2018 – June 2022

PROFESSIONAL EXPERIENCE

Data Annotation Freelance AI Consultant AI Trainer in Bioinformatics, Data Science and Healthcare - Design bioinformatics and ML tasks in transcriptomics and epigenetics to assess AI agents on problem-solving. - Generate the gold-standard solution, including inputs, context, code, plots, and written reports for each bioinformatics task. - Evaluate LLM outputs (3+ models) against the gold-standard solution using custom atomic and self-contained rubrics to generate feedback for reinforcement learning.	Vancouver, Canada July 2026 - Present
BC Cancer Research Institute Research Assistant Birol Lab <i>Simulation of Haplotype-Resolved Single-Cell Methylation Profiles</i> - Simulated haplotype-resolved, cell-type-specific single-cell methylation profiles using an overdispersed generalized linear mixed model, and ensured reproducible execution with Snakemake. - Performed <i>in silico</i> bisulfite conversion, paired-end reads simulation, alignment, methylation call extraction. - Assessed sequencing depth effects on CpG methylation detection, achieving a mean absolute error of 0.031, 98% CpG recovery, and an adjusted Rand index of 0.70 for cell-type classification by hierarchical clustering. <i>AIEdit: Alignment-free genome assembly polisher trained on spaced seed match patterns</i> (PLOS Computational Biology) - Co-authored to AIEdit, a platform-agnostic, alignment-free genome assembly polisher that uses spaced-seed match patterns and a GRU-based neural network to correct base-level errors in long-read assemblies. - Validated polished human and fruit fly de novo assemblies using QUAST for mismatch/indel error rates. - Measured polishing accuracy, runtime, and memory, improving human assembly QV from 28.7 to 32.9 in 9.5 hours.	Vancouver, Canada Sep. 2024 – Sep. 2026
BBVA S.A Data Scientist Associate COE, Enterprise & Government Banking - Developed a product engagement metric based on usage patterns, increasing product usage by 30%. - Developed a classification model to identify credit-eligible clients and estimate credit amounts. - Served as a stakeholder in two scrum projects, contributing to planning, management, and solution development.	Mexico City, Mexico May 2023 - July 2024
Data Scientist Analyst Enterprise & Government Banking - Trained 14 statistical and deep learning time-series models (e.g., ARIMA, ETS-LSTM) to forecast 90-day income, expenses, and balances. Ensembled the best three models. Effectively predicted 80% of clients' amounts. - Implemented unit testing for production engines in PySpark, ensuring code reliability and quality standards.	Oct. 2022 - May 2023

SKILLS

Programs: Python, R, PySpark, Bash. **AI Tools:** Claude, Codex. **Tools:** Git, GitHub, Snakemake, Nextflow, Docker, HPC, SLURM. **Bioinformatics:** Single-Cell Analysis, Machine Learning, Deep Learning, Epigenetics, Transcriptomics, Genomics.

ADDITIONAL EXPERIENCE

Canadian Epigenetics, Environment & Health Research Consortium (CEEHRC) Podcast Team Member & CEEHRC Seminars Committee Representative - Produce podcast episodes featuring interviews with Canada-based researchers about their career in epigenetics (Season 4). - Attend monthly meetings to discuss seminar topics, speakers, and upcoming scientific events.	Vancouver, Canada Jun. 2025 - Present
Intelligence Trade, S.L Moondi <i>Intern in Mathematical Modeling and Data Validation</i> - Identified key macroeconomic and demographic factors influencing Spanish wine export companies. - Ranked the 20 ideal countries for wine exports from Spain based on wine type, costs, geospatial factors.	Mexico City, Mexico Feb. 2022 - Aug. 2022
Colegio Nacional de Matemáticas (CONAMAT) <i>Teacher of Mathematics</i> - Supported students to gain admission to high-prestige Mexican universities. - Delivered lessons at various school levels to groups ranging in size from 10 to 40 students.	Mexico City, Mexico Apr. 2016 - May 2019

Bioconductor - BioNA Hackathon 2026

Seattle, USA

Benchmarking LLMs Bioinformatics Produced Workflows

Aug. 2026

- Developed *llmbiobench*, a proof-of-concept R package to benchmark LLM-driven workflows (20+ tasks).
- Identified gaps in AI-generated code reproducibility and tool selection in agents (STELLA, Biomni, Claude, and Gemini).

The University of British Columbia (UBC)

Vancouver, Canada

Unravelling the Complexity of Medulloblastoma Group 3 and 4

Jan. 2025 - Apr. 2025

- Analyzed medulloblastoma scRNA-seq data to compare Group 3 and Group 4 tumours using QC, normalization, differential expression, pathway enrichment, and unsupervised clustering (k-means, k-medoids, and hierarchical clustering).
- Characterized immune cell composition across subgroups and clusters, identifying potential immunotherapy opportunities.

Recurrence-Free Survival Modelling in Low-Grade Ta Bladder Cancer

Mar. 2025

- Built a LASSO-penalized Cox proportional hazards model to predict recurrence-free survival in low-grade, stage Ta bladder cancer, combining clinical variables with highly variable genes (8,000 +) from transcriptomic data.
- Trained on the UROMOL cohort and externally validated on an independent cohort, evaluating performance with ROC (3 months to 5 years) and Kaplan–Meier curves for risk groups.

Wikipedia webpage: “*Pan-genome graph construction*”

March. 2025 - Apr. 2025

- Wikipedia article “*Pan-genome graph construction*”, covering graph-based methodologies and applications in personalized medicine and population genomics [[wikipedia.org/wiki/Pan-genome graph construction](https://wikipedia.org/wiki/Pan-genome_graph_construction)].

Benchmarking Single-Cell Classification Models in the Mouse Brain

Jan. 2025 - Apr. 2025

- Applied cell and gene quality control, filtering, and doublet removal to Tabula Muris mouse brain scRNA-seq data.
- Benchmarked machine learning models (SVM, CellTypist) and deep learning/single-cell annotation models (scANVI, Cell BLAST) across brain populations, achieving accuracy and F1 scores of 0.98, and visualized results with t-SNE.

Automated Single Nucleotide Variant Calling Pipeline for Population-Scale Genomic Analysis

Sep. 2024 - Dec. 2024

- Engineered a Nextflow and Docker pipeline for genome alignment, PCR duplicate removal, and SNVs identification.
- Validated the workflow through end-to-end variant discovery on raw *Drosophila melanogaster* paired-end WGS datasets.

Low-temperature Stress Alters Kidney Transcriptome in Takifugu rubripes

Sep. 2024 - Dic. 2024

- Analyzed kidney RNA-seq data from Takifugu rubripes (Japanese pufferfish) at 5°C and 23°C using STAR aligner and DESeq2, identifying differentially expressed genes associated with cold stress response.
- Performed enrichment analysis with topGO, revealing energy conservation via dysregulated lipid and amino acid metabolism, and upregulation of genes for membrane integrity, circadian rhythm, and apoptosis.

Universidad Nacional Autónoma de México (UNAM)

Mexico City, Mexico

Cost-effectiveness analysis of Paxlovid treatment to reduce severe COVID-19 disease

Oct. 2022 - Aug. 2024

- Simulated COVID-19 patient cohorts and modelled 28-day disease progression with a Markov model.
- Evaluated the cost-effectiveness of guideline-based versus expanded Paxlovid use using Cox model and sensitivity analyses.
- Collaborate with Dr. Yanink N., researcher at National Institute of Medical Sciences and Nutrition Salvador Zubirán (INCMNSZ) and a multidisciplinary team (two medical doctors and a data scientist).

Prediction of the cleavage site of the HIV-1 protease.

Aug. 2021 - Nov. 2021

- Applied feature engineering to extract physicochemical properties from protein structure data.
- Predicted the HIV-1 protease cleavage site using LR, SVM, and Naive Bayes, achieving 90% accuracy.